Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

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Project Overview

Dataset Scale

3,900 purchases across 18 columns with comprehensive customer and transaction data

Strategic Goal

Uncover spending patterns, customer segments, and product preferences for business decisions

Analysis Approach

Python for data preparation, SQL for business questions, Power BI for visualization

This comprehensive analysis combines customer demographics (Age, Gender, Location, Subscription Status) with purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color) and shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type). The dataset contained 37 missing values in Review Rating that were imputed using median ratings by product category.

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Data Preparation & Cleaning

Python Foundation

- Imported dataset using `pandas`
- Initial exploration with `df.info()` and `.describe()`
- Standardized columns to snake case for readability
- Created age_group and purchase_frequency_days features

Quality Assurance

- Imputed 37 missing review ratings using median by category
- Verified and dropped redundant promo_code_used column
- Connected to PostgreSQL for SQL analysis
- Loaded cleaned DataFrame into database

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Revenue by Gender

$157,890

Male Customers

Generated 68% of total revenue

$75,191

Female Customers

Generated 32% of total revenue

2.1x

Revenue Ratio

Male customers spent 2.1x more than female customers

Male customers generated significantly higher revenue compared to female customers, representing a key demographic for strategic focus.

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Product Performance Insights

1

Top-Rated Products

Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.78) received highest average ratings

2

Discount-Heavy Products

Hat (50%), Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), Pants (47.37%) most frequently discounted

3

Best-Selling by Category

Clothing: Blouse & Pants (171 each). Accessories: Jewelry & Sunglasses (161 each). Footwear: Sandals (160). Outerwear: Jacket (163)

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Shipping & Subscription Analysis

Express Shipping

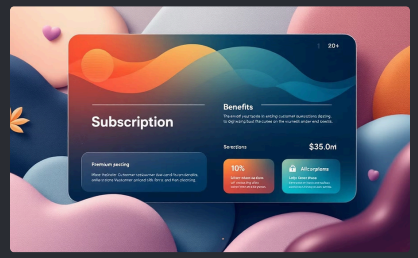
Average purchase: $60.48

Higher-value transactions

Standard Shipping

Average purchase: $58.46

Most common option



Subscription Status

1,053 subscribers (27%) vs. 2,847 non-subscribers (73%)

Express shipping customers spent slightly more ($60.48 vs. $58.46 standard). Non-subscribers generated higher total revenue ($170,436 vs. $62,645 for subscribers) due to larger customer base, though average spend was nearly identical ($59.87 vs. $59.49).

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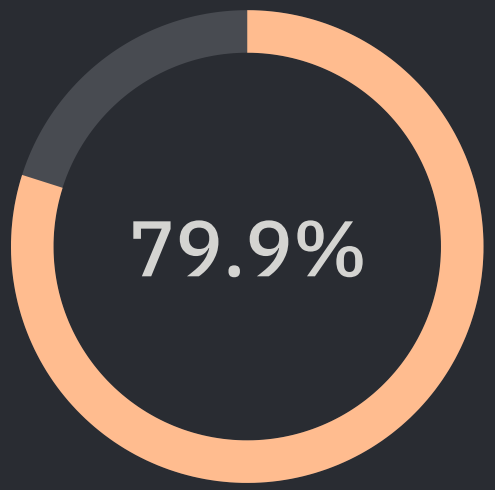
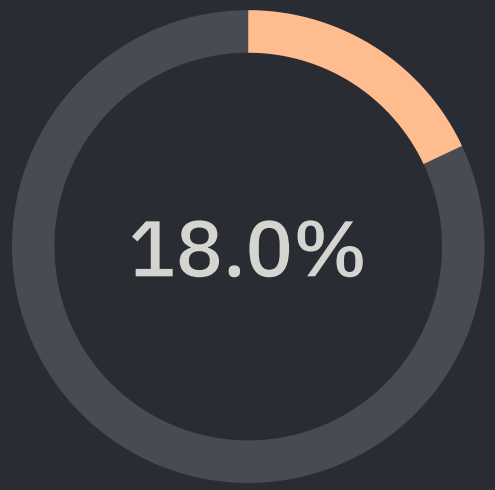
Customer Segmentation



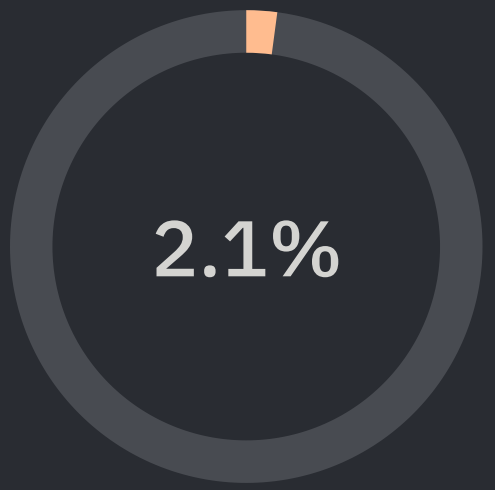
Loyal Customers

3,116 customers with consistent purchase history



Returning Customers

701 customers making second or third purchases



New Customers

83 customers with single initial purchase

The customer base is dominated by loyal customers (79.9%), indicating strong retention. However, new customer acquisition represents only 2.1% of the base, suggesting opportunities for growth in this segment.

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Revenue by Age Group

25.6%

Young Adult

$62,143 total revenue

24.4%

Middle-aged

$59,197 total revenue

23.1%

Adult

$55,978 total revenue

23.0%

Senior

$55,763 total revenue

Revenue distribution across age groups is remarkably balanced, with Young Adults leading at $62,143 (25.6%) and Seniors at $55,763 (23.0%). This even spread suggests broad appeal across demographics.

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Key Findings Summary

High-Value Discount Users

839 customers used discounts but spent above average purchase amount ($60)

Repeat Purchase Correlation

958 repeat buyers (over 5 purchases) vs. 2,518 single-purchase customers

Average Metrics

$59.76 average purchase, 3.75 average rating, 3,900 total customers analyzed

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Strategic Recommendations

01

Boost Subscriptions

Promote exclusive benefits and incentives to increase subscriber base from current 27%

02

Enhance Loyalty Programs

Reward repeat buyers to accelerate movement into loyal customer segment

03

Optimize Discount Strategy

Balance sales boosts with margin control, especially for frequently discounted products

04

Highlight Top Performers

Feature top-rated and best-selling products in marketing campaigns

05

Target High-Value Segments

Focus marketing on male customers and express-shipping users with higher purchase values

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